Technical Approach Report

1. Architectural Overview & Model Choices

For this Proof of Concept (PoC), the architecture was designed with a focus on rapid prototyping, low operational overhead, and high accuracy in natural language understanding.

- **Frontend Interface:** Streamlit was selected as the application framework. It allows for the rapid development of internal data dashboards without the complexity of maintaining separate frontend and backend codebases, making it the ideal choice for deploying internal HR tools.
- **AI Inference Engine**: Google GenAI (gemini-3.8-flash) was chosen as the core Large Language Model (LLM). This model provides excellent zero-shot reasoning capabilities, which is critical for parsing highly unstructured data (like varied candidate resume formats) and returning structured, predictable insights.

2. Data Ingestion & Processing Pipeline

The data pipeline is designed to be seamless, requiring zero technical intervention from the HR end-user.

- **Ingestion:** The recruiter uploads a candidate's resume in a standard PDF format via the secure web interface.
- **Extraction:** The system utilizes the PyPDF2 library in Python to parse the document. It extracts raw textual data while efficiently stripping away complex formatting, tables, or graphical elements that are unnecessary for skill evaluation.
- **Evaluation:** The sanitized text string is securely passed to the Gemini API endpoint alongside the system prompt.
- **Delivery:** The LLM processes the text and returns a structured evaluation, which the application instantly renders into a clean, readable Markdown summary for the recruiter.

3. Prompt Engineering Strategy

To prevent AI hallucinations and ensure highly consistent outputs across hundreds of resumes, the system prompt utilizes a strict "Persona-Based Constraint Structure".

- **System Persona:** The LLM is explicitly instructed to act as an "Internal AI Admissions Evaluator for Cynbit Technologies."
- **Context Grounding:** The prompt injects Cynbit’s specific BitElevate curriculum tracks (Software Engineering, Digital Marketing & SEO, Cloud) directly into the context window, forcing the AI to only recommend these existing pathways.
- **Output Constraints**: The model is mathematically restricted to outputting exactly four data points: Candidate Level, Recommended Track, Brief Reasoning (limited to a maximum of two sentences to ensure brevity), and a quantifiable Match Score (out of 100).

4. Execution Plan & Production Scaling Strategy

While the current PoC successfully validates the automated screening logic, deploying this to a production environment handling thousands of applications requires targeted architectural upgrades.

- **Resolving API Latency:** The current reliance on free-tier APIs introduces intermittent network latency during the evaluation phase. For production, the backend inference engine will be migrated to the Groq API (utilizing Llama-3 models running on specialized Language Processing Units). This hardware-level upgrade will reduce inference times to sub-second levels, ensuring real-time performance.
- **Database Integration**: The subsequent execution phase involves integrating a vector database (such as ChromaDB) to store processed candidate profiles. This will allow the HR team to dynamically search past applicants using semantic skill matching (e.g., "find candidates with Python and AWS experience") rather than traditional keyword searches.